

Fuel Leakage Switch Signal Malfunction

Symptoms

- Original equipment manufacturer (OEM) fuel leakage switch has malfunctioned.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The fuel leakage switch is a normally closed switch. The alarm sounds when there is an open in the circuit.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box (C.I.B.) wiring.	
	STEP 1A. Check the fuel leakage SIGNAL and RETURN wires for an open circuit.	Greater than 100k ohms?
	STEP 1B. Check the fuel leakage SIGNAL and RETURN wires for a wire-to-wire short circuit.	Less than 10 ohms?
	STEP 1C. Check the fuel leakage SIGNAL wire for a short circuit to ground.	Less than 10 ohms?
STEP 2.	Check OEM wiring harness.	
	STEP 2A. Check the fuel leakage SIGNAL and RETURN wires for an open circuit.	Greater than 100k ohms?

STEPS		SPECIFICATIONS
	STEP 2B. Check the fuel leakage SIGNAL and RETURN wires for a wire-to-wire short circuit.	Less than 10 ohms?
	STEP 2C. Check the fuel leakage SIGNAL wire for a short circuit to ground.	Less than 10 ohms?

STEP 1. Check the customer interface box (C.I.B.) wiring.

STEP 1A. Check the fuel leakage SIGNAL and RETURN wires for an open circuit.

Conditions : • Open the C.I.B. • Disconnect the fuel leakage SIGNAL and RETURN wires at the control panel. • Disconnect the C2 connector.		
Action	Specification/Repair	Next Step

Check the fuel leakage SIGNAL and RETURN wires for an open circuit. Note: An alarm will sound on the remote input/output unit when an open circuit is detected. - Place one test lead on the fuel leakage SIGNAL wire at the control panel. - Place the other test lead on the fuel leakage SIGNAL pin at the C2 connector. - Place one test lead on the fuel leakage RETURN wire at the control panel. - Place the other test lead on the fuel leakage RETURN pin at the C2 connector. Refer to the circuit diagram or the wiring diagram for connector pin identification.	Greater than 100k ohms? YES	1B
	Greater than 100k ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete

STEP 1B. Check the fuel leakage SIGNAL and RETURN wires for a wire-to-wire short circuit.

Conditions : - Open the C.I.B. - Disconnect the fuel leakage SIGNAL and RETURN wires at the control panel. - Disconnect the C2 connector.		
Action	Specification/Repair	Next Step

Check the fuel leakage SIGNAL and RETURN wires for a wire-to-wire short circuit. • Place one test lead on the fuel leakage SIGNAL wire at the control panel. • Place the other test lead on all other pins at the control panel. • Place one test lead on the fuel leakage RETURN wire at the control panel. • Place the other test lead on all other pins at the control panel. Refer to the circuit diagram or the wiring diagram for connector pin identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
	Less than 10 ohms? NO	1C

STEP 1C. Check the fuel leakage SIGNAL wire for a short circuit to ground.

Conditions : • Open the C.I.B. • Disconnect the fuel leakage SIGNAL wire at the control panel. • Disconnect the C2 connector.		
Action	Specification/Repair	Next Step

Check the fuel leakage SIGNAL wire for a short circuit to ground. - Place one test lead on the fuel leakage SIGNAL wire at the control panel. - Place the other test lead on the panel ground. Refer to the circuit diagram or the wiring diagram for connector pin identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
	Less than 10 ohms? NO	2A

STEP 2. Check the OEM wiring harness.

STEP 2A. Check the fuel leakage SIGNAL and RETURN wires for an open circuit.

Conditions :

- Disconnect the OEM harness at the C2 and C9 connectors.
- Disconnect the fuel leakage sensor connector.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Check the fuel leakage SIGNAL and RETURN wires for an open circuit.

Note: An alarm will sound on the remote input/output unit when an open circuit is detected.

Greater than 100k ohms?

YES

2B

- Place one test lead on the fuel leakage SIGNAL pin at the C2 connector.
- Place the other test lead on the fuel leakage SIGNAL pin at the C9 connector.
- Place one test lead on the fuel leakage RETURN pin at the C2 connector.
- Place the other test lead on the fuel leakage RETURN pin at the C9 connector.
- Place one test lead on the fuel leakage SIGNAL pin at the C9 connector.
- Place the other test lead on the fuel leakage SIGNAL pin at the sensor connector.
- Place one test lead on the fuel leakage RETURN pin at the C9 connector.
- Place the other test lead on the fuel leakage RETURN pin at the sensor connector.

Refer to the circuit diagram or the wiring diagram for connector pin identification.

Greater than 100k ohms?

NO

Repair :

Replace the wire. Refer to Procedure 015-023 in Section 15.
(/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)

Repair complete

STEP 2B. Check the fuel leakage SIGNAL and RETURN wires for a wire-to-wire short circuit.

Conditions :

- Disconnect the OEM harness at the C2 and C9 connectors.
- Disconnect the fuel leakage sensor connector.

Action	Specification/Repair	Next Step
Check the fuel leakage SIGNAL and RETURN wires for a wire-to-wire short circuit. • Place one test lead on the fuel leakage SIGNAL pin at the C2 connector. • Place the other test lead on all other pins at the C2 connector. • Place one test lead on the fuel leakage SIGNAL pin at the C9 connector. • Place the other test lead on all other pins at the C9 connector. Refer to the circuit diagram or the wiring diagram for connector pin identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
	Less than 10 ohms? NO	2C

STEP 2C. Check the fuel leakage SIGNAL wire for a short circuit to ground.**Conditions :**

- Disconnect the OEM harness at the C2 and C9 connectors.
- Disconnect the fuel leakage sensor connector.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Check the fuel leakage SIGNAL wire for a short circuit to ground. - Place one test lead on the fuel leakage SIGNAL pin at the C2 connector. - Place the other test lead on the panel ground. Refer to the circuit diagram or the wiring diagram for connector pin identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
	Less than 10 ohms? NO Repair : Replace the fuel leakage switch. Contact a Cummins® Authorized Repair Location. Refer to Procedure 019-150 in Section 19. (/qs3/pubsys2/xml/en/procedures/300/300-019-150.html)	Repair complete